

Lab 01: Installing WampServer and Setting Up the MySQL Database Environment

Objective:

The objective of this lab is to install WampServer on a Windows machine and use it to set up a local MySQL database server. By the end of this lab, the student should be able to install WampServer, start its services, and access the MySQL database through phpMyAdmin to prepare the environment for future database exercises.

Introduction:

WampServer is a Windows-based web development platform that bundles the Apache web server, the PHP scripting language, and a MySQL/MariaDB database server into a single, easy-to-install package. It also includes phpMyAdmin, a web-based graphical tool used to create, manage, and query MySQL databases without typing raw SQL commands from the terminal.

For a Database Systems course, WampServer provides a convenient way to run a real MySQL server locally, without needing an internet connection or a remote server. This lab focuses on downloading and installing WampServer, starting its services, and confirming that the MySQL database engine and phpMyAdmin interface are working correctly.

Procedure:

1. Open a web browser and go to the official WampServer download page (wampserver.aviatechno.net).

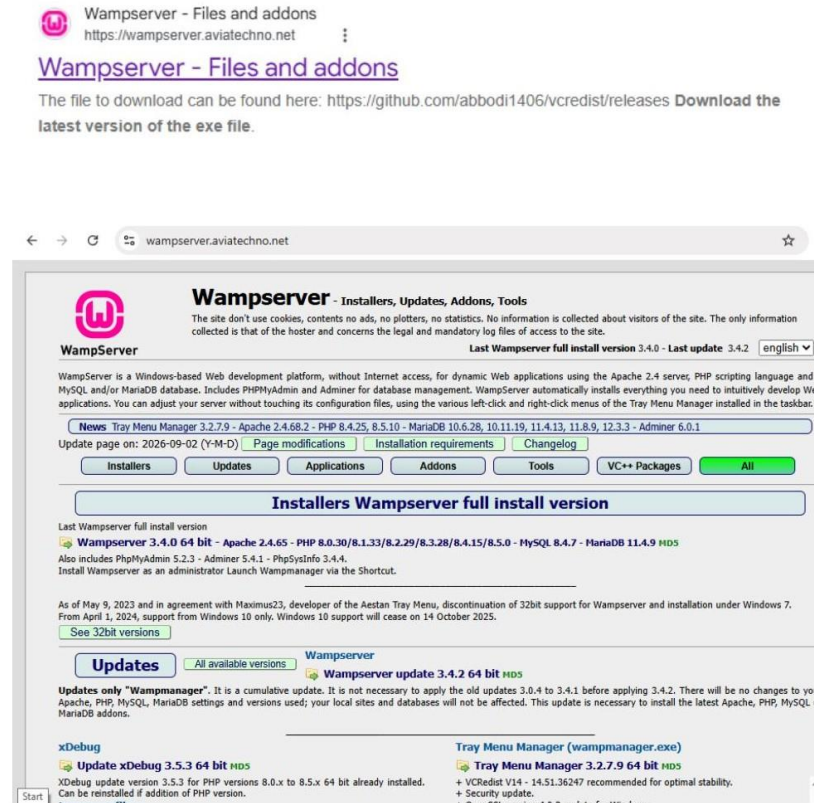


Figure 1: WampServer official download page showing the full install version.

2. Click the Installers/download button and download the WampServer full install version (e.g., wampserver3.4.0_x64.exe). Wait for the download to complete.

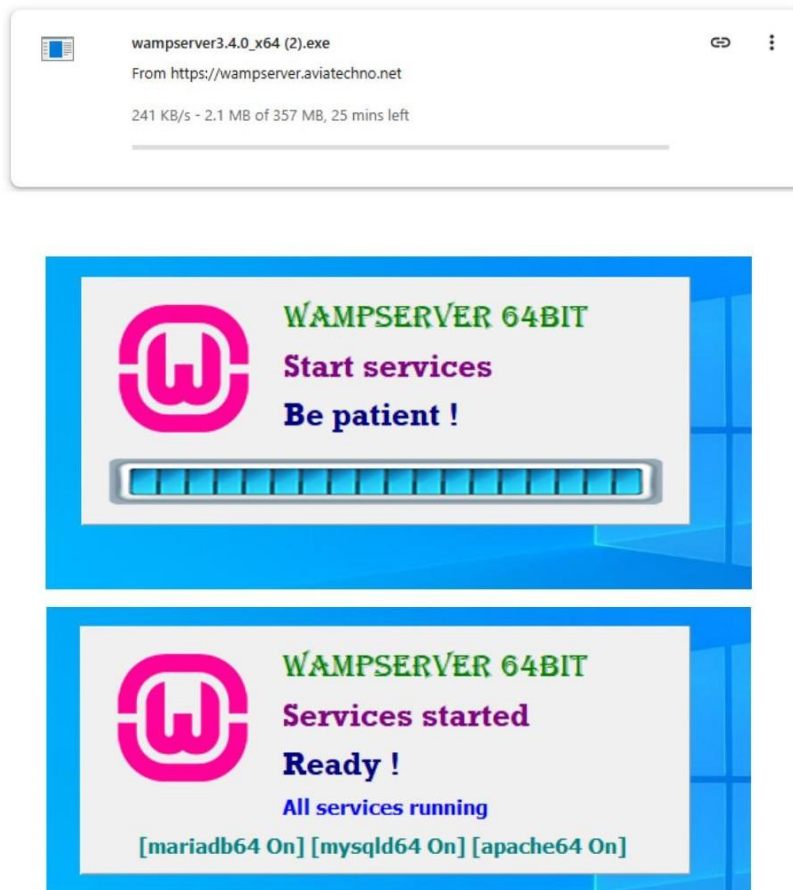


Figure 2: WampServer installer file downloading in the browser.

3. Run the downloaded .exe file as Administrator and follow the installation wizard using the default settings (this installs Apache, MySQL/MariaDB, and PHP together).
4. After installation, launch WampServer. The tray manager will display a loading screen while it starts all services (Apache, MySQL, MariaDB) wait for this process to finish.
5. Once all services are running, WampServer confirms this with a 'Services started Ready!' message, showing MariaDB, MySQL, and Apache all set to On.

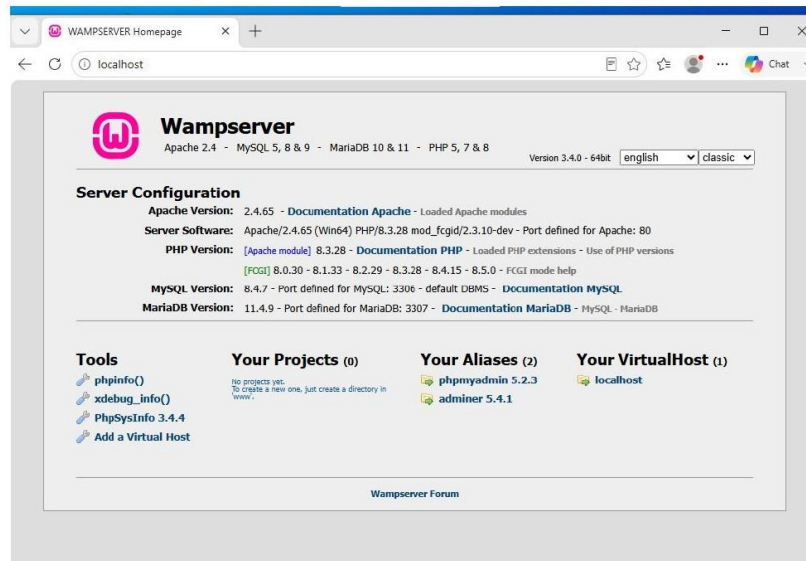


Figure 3: WampServer homepage (localhost) showing server configuration Apache, PHP, and MySQL/MariaDB versions confirming all services are running.

6. Open a browser and navigate to localhost to confirm the WampServer homepage loads, displaying the installed Apache, PHP, and MySQL/MariaDB versions.
7. From the WampServer homepage, click on phpMyAdmin under 'Your Aliases' to open the database management tool.
8. On the phpMyAdmin login page, enter the username 'root' and leave the password field blank (default configuration), then click Log in.

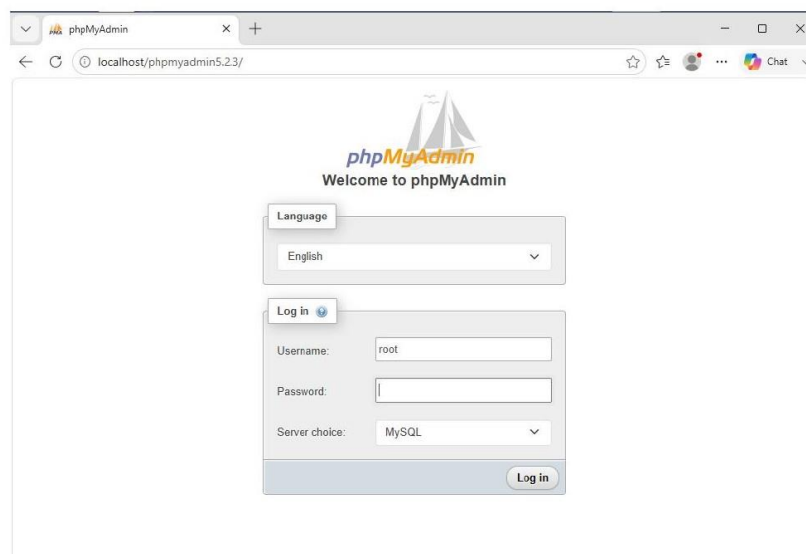


Figure 4: phpMyAdmin login page with default root credentials.

Result:

WampServer was successfully downloaded and installed on the system. After launching the application, all required services Apache, MySQL, and MariaDB started correctly, as confirmed by the 'All services running' status and the WampServer homepage displaying the server configuration details. Logging into phpMyAdmin with the default root account was successful, confirming that the MySQL database engine is accessible and ready to be used for creating and managing databases in upcoming labs.

Conclusion:

This lab provided hands on experience in setting up a local database development environment using WampServer. By downloading, installing, and configuring WampServer, a working MySQL server was established along with phpMyAdmin as a graphical interface for database management. This environment forms the foundation for future Database Systems labs, which will involve creating databases, tables, and executing SQL queries.